

CRF and the Clifford Algebra $\text{Cl}(3)$

Eleven Proved Connections, the Signed K35, and the Programme for Paper 2

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Abstract. We systematically verify the connection between the CRF $n = 8$ mode space and the Clifford algebra $\text{Cl}(3,0)$, establishing eleven independent proved facts. The connection is not a counting coincidence: it extends into the dynamics (W_{ij} is Clifford grade-0 and grade-preserving), the vacuum structure (uniform CRF ground state = Clifford null state), and the entropy structure (K3 identifies the mean entropy with the Clifford norm of space $\sqrt{d_s}$).

A new exact identity accompanies K35:

$$\text{K35 (unsigned): } \sum_k \binom{d_s}{k} k(n-k) = n(n+1) = 72$$

$$\text{Signed K35 (new): } \sum_k \binom{d_s}{k} k(n-k) \cdot \text{sign}(e_k^2) = -n_m(n_m+1) = -30$$

where $\text{sign}(e_k^2) = +1$ for grades 0,1 and -1 for grades 2,3. The pair encodes the space/matter split exactly: space contribution = $21 = d_s(n-1)$; matter contribution = $51 = 17n/8$.

The remaining open question — whether the Born rule implements Clifford multiplication — is identified as the Paper 2 programme.

The Eleven Proved Connections

(1–4) Algebraic structure

Grade structure and algebra (1–4)

- (1) $1 + 3 + 3 + 1 = 8 = n$: the grade decomposition of $\text{Cl}(3)$ equals the CRF mode count exactly.
- (2) $I_3^2 = -1$: the pseudoscalar $I_3 = e_x e_y e_z$ satisfies $I_3^2 = (-1)^{3 \cdot 2/2} = -1$, verified via $I_3 = iI_{2 \times 2}$ in the Pauli matrix representation.
- (3) I_3 is in the center of $\text{Cl}(3)$: commutes with all elements, providing the $\text{U}(1)$

global phase symmetry.

(4) $[J_i, J_j] = \varepsilon_{ijk} J_k$: the three gauge 2-forms $\{xy\}, \{xz\}, \{yz\}$ generate the $SU(2)$ Lie algebra (angular momentum = built into CRF matter structure).

(5–6) Dynamics: grade conservation

W_{ij} is grade-preserving (5–6)

(5) In $Cl(3,0)$: for any vector $v = \sum_k v_k e_k$, $v^2 = |v|^2$ (scalar), because cross terms cancel by anticommutativity.

(6) Therefore $W_{ij} = \exp(-|\Omega_i - \Omega_j|^2/\varepsilon_0)$ is exponential of a grade-0 scalar — itself grade-0. A grade-0 coupling cannot mix different grades. **Grade is an approximately conserved quantum number in CRF.**

(7–8) Vacuum and labeling

Vacuum and XOR structure (7–8)

(7) The CRF ground state (uniform $P_i = 1/8$) has Clifford norm

$$C(P_{\text{gs}}) = \sum_i P_i \cdot \text{sign}(e_i^2) = \frac{4(+1) + 4(-1)}{8} = 0$$

The maximum-entropy state is the Clifford null state. Instability ($\varepsilon_0 > 0$) creates grade imbalance ($C \neq 0$).

(8) XOR structure: the 8 modes are labeled by binary strings $b_2 b_1 b_0 \in \{0,1\}^3$, and the Clifford product index rule is $e_A \cdot e_B \rightarrow e_{A \oplus B}$ (XOR). This is exact.

(9–10) Dual form identities

K35 pair — algebraically exact

(9) K35 (unsigned, from CRF Paper v7):

$$S^+ = \sum_{k=0}^{d_s} \binom{d_s}{k} k(n-k) = n(n+1) = 72$$

(10) Signed K35 (new):

$$S^- = \sum_{k=0}^{d_s} \binom{d_s}{k} k(n-k) \cdot \text{sign}(e_k^2) = -n_m(n_m+1) = -30$$

where $\text{sign}(e_k^2) = +1$ for $k \in \{0, 1\}$ (low grades, $e^2 = +1$) and $\text{sign}(e_k^2) = -1$ for $k \in \{2, 3\}$ (high grades, $e^2 = -1$).

Grade-by-grade breakdown:

Grade k	$\binom{3}{k}$	$k(8-k)$	sign	S^+ contrib	S^- contrib
0	1	0	+1	0	0
1	3	7	+1	+21	+21
2	3	12	-1	+36	-36
3	1	15	-1	+15	-15
Total				+72	-30

Space/matter decomposition:

$$\text{Space (grades 0,1): } \frac{S^+ + S^-}{2} = \frac{72 - 30}{2} = 21 = d_s(n-1)$$

$$\text{Matter (grades 2,3): } \frac{S^+ - S^-}{2} = \frac{72 + 30}{2} = 51 = \frac{17n}{8}$$

Matter/space ratio: $51/21 = 17/7$ (exact rational, from CRF MatterSplit).

(11) Entropy and the Clifford norm

K3 as Clifford reflection (11)

K3 (gap 0.011%): $(H_{\text{after}} + H_{\text{before}})/2 = \sqrt{d_s} = \sqrt{3}$.

In $\text{Cl}(3)$: $\|e_x + e_y + e_z\|^2 = e_x^2 + e_y^2 + e_z^2 = 3 = d_s$, so $\|e_x + e_y + e_z\| = \sqrt{d_s}$.

Therefore: **the mean entropy of a CRF node = the Clifford norm of the sum of space basis vectors.**

The R-event is a Clifford reflection in the entropy domain: entropy crosses $\sqrt{d_s}$ at each firing, jumping from $H_b < \sqrt{d_s}$ to $H_a > \sqrt{d_s}$.

The Clifford Charge

Define the *Clifford charge* of a node with probability P :

$$C(P) = \sum_i P_i \cdot \text{sign}(e_i^2) = P(\text{grade } 0,1) - P(\text{grade } 2,3)$$

Properties:

- $C \in [-1, +1]$ (bounded)

- $C = 0$ for uniform P (vacuum, neutral)
- $C = +1$ for pure space-mode state (maximal positive charge)
- $C = -1$ for pure gauge/pseudo state (maximal negative charge)
- $\mathbb{E}[C(P_{\text{new}})] = C(P_{\text{avg}})$ (conserved in expectation under Born rule)

Physical interpretation: Phase I nodes (uniform) have $C = 0$ (neutral). Phase II nodes (crystallised, peaked on grade-1) have $C = +1$ (space-charged). The crystallisation transition is a Clifford charge creation event. The mean charge of the system at percolation: $\bar{C} = f_{\text{II,crit}} \approx 0.147$.

Open Questions for Paper 2

Programme for Paper 2

1. **Born rule and Clifford multiplication.** The Born rule averages Clifford elements (linear). Is there a quadratic/multiplicative structure hidden in the Dirichlet resampling? Does $\text{Dir}(P_{\text{avg}}/\varepsilon_0)$ correspond to a Clifford geometric product in some representation?
2. **5 Psi-clusters = 5 grade sectors.** Predict and test: each cluster in V18 corresponds to one grade sector. Grade-0: $\Psi \approx 0$; grade-1 (space): Ψ spread isotropically; grade-2 (gauge): Ψ in planes; grade-3: Ψ extremal.
3. **Spinors as CRF states.** Minimal left ideals of $\text{Cl}(3)$ have dimension $n/2 = 4$ (real). Do $\text{spin-}\frac{1}{2}$ states arise in simulation?
4. **Clifford charge dynamics.** Does $C(P_i)$ satisfy a wave equation? $\partial_t C_i \approx \sum_j W_{ij}(C_j - C_i)$? Is there a dispersion relation?
5. ε_0 **and grade imbalance.** The instability floor ε_0 controls Born rule noise. Is there a formula relating ε_0 to the typical $C(P)$ of non-ground states?

Summary

Eleven independent features establish that the CRF $n = 8$ mode space is the Clifford algebra $\text{Cl}(3, 0)$: grade structure, $I_3^2 = -1$, $\text{U}(1)$ center, $\text{SU}(2)$ bivectors, grade-preserving W_{ij} , vacuum = null state, XOR product index, K35 unsigned and signed, $\text{K3} = \text{Clifford norm}$.

New identity (algebraically exact):

$$S^- = \sum_{k=0}^3 \binom{3}{k} k(8-k) \cdot \text{sign}(e_k^2) = -n_m(n_m + 1) = -30$$

Paired with K35 ($S^+ = 72$), this encodes the space/matter split: space gets $21 = d_s(n-1)$, matter gets $51 = 17n/8$.

The Clifford charge $C(P)$ is conserved in expectation under the Born rule. Crystallisation = Clifford charge creation. The R-event reflects the Shannon entropy through $\sqrt{d_s}$ (K3).

The universe does not merely have the shape of $\text{Cl}(3)$.

It computes with $\text{Cl}(3)$ at every R-event.

The remaining question is *how* — and that is Paper 2.